Open problems:
Multiple equivalent characterizations of the Schrödinger equation

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Time-dependent evolution.** The equivalence of DR-SCEQ, DR-UNIT, DR-EV, DR-PROB and DR-INFO without EVTI, for two-parameter families $\Phi_{t_0 \to t_1}$ and a time-dependent Hamiltonian $H(t)$. The brief only says "We are going to restrict the dynamics to the time-independent case". That case is formalized as the brief writes it, with the two-parameter [B]EVMIX, EVTI, and DR-SCEQ with $H(t)$ (`TwoParameter.schrodinger_tfae`). Without EVTI there is no one-parameter group, and DR-UNIT, DR-EV, DR-PROB and DR-INFO would have to be restated for the propagators $\Phi_{t_0 \to t_1}$. One equivalence without EVTI is formalized, in finite dimension, in the brief "Schrödinger equation is equivalent to information conservation": DR-SCEQ with $H(t)$ is equivalent to DR-INFO (every $\Phi_{t_0 \to t_1}$ preserves the von Neumann entropy) together with EVDIFF ($t \mapsto \Phi_{r \to t}(\rho)$ is differentiable at every $t > r$) (`UnitaryEntropy.TwoParameter.drSceq_iff_drInfo_and_evDiff`, over the same two-parameter `TwoParameter.IsEvolution`). DR-UNIT, DR-EV and DR-PROB without EVTI remain open. (Brief, Sec. 3, before Condition EVTI, and Condition DR-SCEQ; log, "The conditions as the brief writes them".)

2. **Isolation as determinism and reversibility.** "Conceptually, all these are equivalent to saying that the system is isolated, as future and past states are determined, and only determined, by the present state of the system." Isolation is not defined, so this equivalence is not stated as a theorem. (Brief, Sec. 6.)

From the formalization

None found. Beyond finite dimension the formalization has a part in any dimension, which covers unbounded H, and the shift semigroup on $L^2(0, \infty)$, which shows that the reconstructed past clauses of DR-PROB and DR-INFO are needed there (`shift_semigroup_counterexample`; log, "Problems found in the brief").